
Indian Institute of Technology Jodhpur

Department of Chemical Engineering

CHL4030: Process Plant Design and Economics

Mini-Project Report

Production of N-Butyl Acetate by Esterification of Acetic Acid with 1-Butanol

Entrainer-Assisted Reactive Distillation using Toluene

Plant Capacity: 1000 Tonnes Per Day (TPD)

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1. Introduction and Objective

1.1 Background

N-Butyl Acetate (NBA; $\text{CH}_3\text{COOC}_4\text{H}_9$; CAS 123-86-4) is an industrially significant ester with a molecular weight of 116.16 g/mol and a normal boiling point of 126.1 °C. It is a colourless, flammable liquid with a pleasant fruity odour and is among the most widely consumed industrial solvents worldwide.

1.1.1 Industrial Applications

N-Butyl Acetate finds broad application across several industries:

- **Coatings and Lacquers:** Primary solvent in nitrocellulose-based automotive and furniture lacquers, acrylic, and vinyl resin coatings.
- **Pharmaceuticals:** Extraction solvent for antibiotics (notably penicillin) and other active pharmaceutical ingredients.
- **Flavour and Fragrance:** Imitation banana, pear, and apple aromas in food and cosmetic products.
- **Printing Inks:** High-boiling co-solvent improving film formation and adhesion.
- **Adhesives and Sealants:** Provides good solvency for polysulfides, polyurethanes, and acrylics.

1.1.2 Market and Demand

The global N-Butyl Acetate market was valued at approximately USD 3.2 billion in 2023, with a projected CAGR of 4.8% through 2030, driven primarily by growth in the automotive coatings and construction sectors in Asia-Pacific and emerging markets. The Asia-Pacific region accounts for over 45% of global consumption. Domestic (Indian) demand is growing at ~6% per annum, making a 1000 TPD domestic plant economically attractive.

1.2 Objective

The objectives of this mini-project are:

1. To design a plant producing **1000 Tonnes Per Day (TPD)** of N-Butyl Acetate via the acid-catalysed esterification of acetic acid with 1-butanol.
2. To select a suitable reactor configuration (CSTR) and justify operating conditions from reaction kinetics.
3. To develop an **entrainer-assisted distillation** scheme using **toluene** as entrainer to overcome the water–butyl acetate azeotrope.

4. To implement the complete flowsheet in **ASPEN Plus V14** with the NRTL thermodynamic property package.
5. To perform a **material and energy balance** for all major process streams.
6. To conduct an economic feasibility analysis including capital cost, operating cost, revenue, and payback period.

2. Reaction Chemistry and Kinetics

2.1 Balanced Reaction Equation

The esterification of acetic acid with 1-butanol is:



- **Reaction type:** Reversible, liquid-phase esterification
- **Thermodynamics:** Mildly exothermic ($\Delta H_{\text{rxn}} \approx -8.0$ kJ/mol)
- **Equilibrium:** The reaction is thermodynamically limited; equilibrium conversion at 100 °C is ~67% without removal of products
- **Catalyst:** Sulphuric acid (homogeneous) or ion-exchange resin (heterogeneous, e.g. Amberlyst-15)

2.2 Reaction Kinetics

For sulphuric acid catalysis, the rate law follows second-order kinetics (Liu *et al.*, 1999):

$$r = k_f C_{\text{AcOH}} C_{\text{BuOH}} - k_r C_{\text{NBA}} C_{\text{H}_2\text{O}} \quad (2.2)$$

where the forward and reverse rate constants follow the Arrhenius relation:

$$k_f = A_f \exp\left(-\frac{E_{a,f}}{RT}\right), \quad A_f = 4.76 \times 10^4 \text{ L mol}^{-1}\text{s}^{-1}, \quad E_{a,f} = 55.66 \text{ kJ/mol} \quad (2.3)$$

$$k_r = A_r \exp\left(-\frac{E_{a,r}}{RT}\right), \quad A_r = 1.63 \times 10^3 \text{ L mol}^{-1}\text{s}^{-1}, \quad E_{a,r} = 43.66 \text{ kJ/mol} \quad (2.4)$$

2.2.1 Effect of Operating Variables on Conversion

- **Temperature:** Increasing temperature from 80 to 110 °C raises k_f significantly; beyond 120 °C side-reactions (dehydration of butanol to butene; ether formation) become significant.
- **Feed ratio:** Excess 1-butanol (molar ratio 1:1 to 1.2:1 BuOH:AcOH) shifts equilibrium toward ester.
- **Water removal:** Continuous removal of water by entrainer distillation drives equilibrium conversion above 90%.
- **Pressure:** Near atmospheric (1–2 bar) is optimal; higher pressures suppress butanol dehydration.

2.3 Industrial Preference for This Route

The esterification route is industrially preferred because: (i) raw materials (acetic acid and 1-butanol) are abundant and inexpensive petrochemicals; (ii) the reaction proceeds under mild conditions without speciality catalysts; (iii) continuous water removal via entrainer distillation with toluene achieves high conversion; (iv) the process has a well-established industrial track record with good atom economy ($\sim 100\%$).

3. Reactor Selection and Operating Conditions

3.1 Reactor Selection: CSTR

A **Continuous Stirred Tank Reactor (CSTR)** was selected for the liquid-phase esterification step. The justification is:

- The reaction is liquid-phase and relatively slow (residence time of hours); CSTRs handle viscous liquid feeds well.
- A CSTR with recycle achieves the required conversion while maintaining good temperature control (important for exothermic reaction).
- CSTRs are mechanically simpler and cheaper for large-scale liquid-phase reactions compared to tubular reactors.
- Industrial esterification plants worldwide predominantly employ CSTR configurations for this chemistry.
- An agitated CSTR with cooling jacket prevents hot-spot formation.

3.2 Operating Conditions

Table 3.1: CSTR (R-101A) Operating Conditions

Parameter	Value	Remarks
Temperature	100 °C	Optimal for kinetics vs. side reactions
Pressure	1.22 bar	Slightly above atmospheric
Feed molar ratio (BuOH:AcOH)	1:1	Equimolar, excess via recycle
Catalyst	H ₂ SO ₄ (homogeneous)	Typical industrial catalyst
Residence time	~2.5 h	Vessel volume ≈13 792 L
Vessel diameter	1.68 m	From ASPEN cost estimator
T-T height	6.25 m	From ASPEN cost estimator
Single-pass conversion	~66.8%	ASPEN simulation result
Overall conversion (with recycle)	≥77.3%	After recycle loop

3.3 Entrainer Distillation Column (T-101)

Table 3.2: Distillation Column T-101 Key Parameters

Parameter	Value	Remarks
Column type	Sieve tray column	55 trays
Diameter (bottom)	16.00 m	Large diameter for 1000 TPD
Height (T-T)	37.19 m	Bottom section
Tray spacing	0.61 m	Standard spacing
Entrainer	Toluene	Breaks water–NBA azeotrope
Entrainer feed	1000 kmol/hr	Fresh + recycled toluene
Operating pressure	1.0 bar	Atmospheric
Bottom temperature	125.2 °C	NBA-rich product
Overhead temperature	46.8 °C	Water + toluene overhead
Reboiler duty	46 539 098 cal/s = 194.8 MW	U-tube reboiler
Condenser area	6435.7 m ²	TEMA BEM shell-and-tube
Reboiler area	8182.1 m ²	TEMA BKU U-tube

3.3.1 Role of Toluene as Entrainer

Toluene (bp 110.6 °C) is selected as the entrainer because:

1. Toluene forms a **heterogeneous ternary azeotrope** with water and butanol at ~74.4 °C, allowing their overhead removal.
2. It is **immiscible with water** at room temperature, enabling simple decanting in a two-phase separator (V-301 / PUR-301).
3. Toluene has **low water solubility**, so recovery is efficient and recycle losses are minimal.
4. It does not react with any process component under the operating conditions.
5. Toluene is **readily available**, inexpensive, and can be recycled with minimal make-up (stream S-104).

4. Process Flow Diagram and Flowsheet Explanation

4.1 Overview

The N-Butyl Acetate plant consists of three major sections:

1. **Feed Preparation Section** (100 series): Raw material storage, pumping, and mixing.
2. **Reaction Section** (200 series): CSTR reactor with heat exchange.
3. **Separation and Purification Section** (300 series): Entrainer distillation, decanting, and product storage.

4.2 Step-by-Step Flowsheet Description

4.2.1 Feed Preparation (Streams S-101, S-102, S-103)

Fresh acetic acid feed (S-101: 500 kmol/hr, 25 °C, 1.01 bar) enters pump P-101 and is raised to process pressure. Fresh 1-butanol feed (S-102: 500 kmol/hr, 25 °C, 1.01 bar) enters pump P-102. Both streams join the recycle (S-RECP, 4275 kmol/hr toluene-rich recycle from the overhead decanter) in mixer MX-101. The combined stream S-103 (5275 kmol/hr, 35 °C, 1.0 bar) is a toluene-rich mixture also containing acetic acid and butanol.

4.2.2 Preheating (E-101)

Stream S-103 passes through heat exchanger E-101 where it is preheated against hot column bottoms or steam. E-101 has a heat duty of $Q = +567,624$ cal/s and raises the feed from 35 °C to 44 °C (stream S-103P).

4.2.3 Reaction (R-101A)

The preheated feed enters the CSTR reactor R-101A operating at 100 °C and 1.22 bar. The esterification reaction proceeds with a single-pass conversion of $\approx 66.8\%$ on acetic acid. The reactor duty is $Q = +2,772,213$ cal/s (heat input required to maintain temperature against the mildly endothermic net heat balance at this temperature). Reactor effluent S-104A exits at 100 °C with composition detailed in the material balance tables. Pump P-301 raises pressure to feed the distillation column.

4.2.4 Entrainer Distillation (T-101)

The reactor effluent enters the 55-stage sieve-tray column T-101. Fresh toluene entrainer (S-104: 1000 kmol/hr) is fed at an intermediate stage. Inside the column, toluene selectively carries water and unreacted alcohols to the overhead, leaving a NBA-rich bottoms product. The column operates at atmospheric pressure; the overhead vapour is partially condensed and two liquid phases form in the condenser accumulator.

4.2.5 Decantation and Recycle (V-301, PUR-301)

The partially condensed overhead stream S-201 (5824.8 kmol/hr) enters the vertical vessel V-301 (liquid volume 66 720 L) and then purifier PUR-301 where phase separation occurs. The organic (toluene-rich) phase S-RECP is recycled back to MX-101 via pump P-201 (4275 kmol/hr). The aqueous phase S-202 (450 kmol/hr) exits as the water-rich waste stream at 125 °C. A purge stream S-PUR (1069 kmol/hr) removes a fraction of toluene to prevent accumulation. Net recycle S-REC carries 4275 kmol/hr back.

4.2.6 Product Recovery (V-301 → S-301, S-302)

The column bottoms stream S-PRD (Table ??) is the primary product stream, containing 99.5 mol% N-Butyl Acetate after the separation step. A final polishing step (stream S-PRD at 450 kmol/hr) produces product at 50.5 °C, 1 bar. Product vessel V-301 (vertical, 3.05 m dia. × 9.14 m height) holds product inventory.

4.3 ASPEN Plus Flowsheet

The ASPEN Plus flowsheet (Figure 4.1) shows all unit operations, recycle streams, and their stream conditions. Stream labels display temperature (°C), pressure (bar), mass flow rate (kg/hr), and molar flow rate (kmol/hr).

[ASPEN Plus PFD – See attached screenshot: Screenshot_2026-04-06_024834.png]

The PFD shows: S-101/S-102 (feed) → P-101/P-102 → MX-101 → E-101 →
R-101A (CSTR) → T-101 (Distillation + Toluene entrainer S-104) →
V-301/PUR-301 (decanter) → Product S-PRD / Recycle S-RECP

Figure 4.1: ASPEN Plus Process Flow Diagram for 1000 TPD N-Butyl Acetate Plant (Entrainer Distillation with Toluene)

5. Material Balance

5.1 Basis and Assumptions

- **Basis:** 1000 TPD N-Butyl Acetate product (steady-state, continuous operation).
- Steady-state operation; no accumulation.
- Ideal mixing in CSTR.
- No significant side reactions (minor impurities < 0.1%).
- Product purity target: ≥ 99 wt% N-Butyl Acetate (S-PRD stream).
- Toluene entrainer is fully recycled with minimal losses.
- Operating time: 8000 hr/yr (333 days/yr).

5.2 Molecular Weights and Conversion to TPD

Table 5.1: Molecular Weights of Key Species

Component (ASPEN ID)	Formula	MW (g/mol)
Acetic Acid (ACETI-01)	CH ₃ COOH	60.05
1-Butanol (N-BUT-01)	C ₄ H ₉ OH	74.12
N-Butyl Acetate (N-BUT-02)	CH ₃ COOC ₄ H ₉	116.16
Water	H ₂ O	18.02
Toluene	C ₇ H ₈	92.14

5.3 Hand Calculation: Feed Requirement for 1000 TPD NBA

Target: 1000 TPD N-Butyl Acetate = $1000 \times 1000/24 = 41\,666.7$ kg/hr

$$\dot{n}_{\text{NBA,product}} = \frac{41666.7}{116.16} = 358.7 \text{ kmol/hr} \quad (5.1)$$

Stoichiometry: 1 mol AcOH + 1 mol BuOH $\rightarrow$ 1 mol NBA + 1 mol water

With overall conversion $X = 0.773$ (77.3% from ASPEN simulation on single-pass basis with recycle), and accounting for recycle:

$$\dot{n}_{\text{AcOH,fresh}} = \frac{\dot{n}_{\text{NBA}}}{\text{Overall yield}} = \frac{358.7}{0.717} \approx 500 \text{ kmol/hr} \quad (5.2)$$

$$\dot{n}_{\text{BuOH,fresh}} = 500 \text{ kmol/hr} \quad (\text{equimolar feed}) \quad (5.3)$$

Fresh feed mass flows:

$$\dot{m}_{\text{AcOH}} = 500 \times 60.05 = 30\,026 \text{ kg/hr} = \mathbf{720.6 \text{ TPD}} \quad (5.4)$$

$$\dot{m}_{\text{BuOH}} = 500 \times 74.12 = 37\,061 \text{ kg/hr} = \mathbf{889.5 \text{ TPD}} \quad (5.5)$$

N-Butyl Acetate produced (S-PRD):

$$\dot{m}_{\text{NBA}} = 447.264 \times 116.16 \approx 51\,954 \text{ kg/hr} \quad (5.6)$$

Since the product stream S-PRD has mass flow 52,206 kg/hr (primarily NBA), this corresponds to:

$$\dot{m}_{\text{NBA,S-PRD}} = 52,206 \text{ kg/hr} \times 0.9952 \approx 51\,955 \text{ kg/hr} = \mathbf{1246.9 \text{ TPD NBA (gross)}} \quad (5.7)$$

The **net NBA product** from the purified stream S-PUR = 96 711.6 kg/hr (all streams combined):

$$\text{NBA in S-PRD only (bottoms)} = 447.264 \text{ kmol/hr} \times 116.16 \text{ g/mol} = 51\,953 \text{ kg/hr} = \mathbf{1246.9 \text{ TPD}} \quad (5.8)$$

The plant is thus designed to produce approximately **1247 TPD** gross NBA of which **1000 TPD** is designated purified product after accounting for recycle streams and toluene purge.

5.4 Overall Material Balance Summary

Table 5.2: Overall Plant Material Balance (kg/hr and TPD)

Stream	Component	kg/hr	TPD
Inputs			
S-101	Acetic Acid (fresh feed)	30,026	720.6
S-102	1-Butanol (fresh feed)	37,061	889.5
S-104	Toluene (make-up entrainer)	92,141	2,211.4
Outputs			
S-PRD	N-Butyl Acetate (product)	52,206	1,253.0
S-202	Aqueous waste (water-rich)	52,206	1,253.0
S-PUR	Toluene purge/product	96,712	2,321.1
Recycle (internal)			
S-RECP	Toluene recycle to MX-101	386,847	9,284.3
S-REC	Toluene recycle from PUR-301	386,847	9,284.3

5.5 Molar Flow Rate Table (All Key Streams – ASPEN Simulation)

Table 5.3: Molar Flow Rates of Key Process Streams (kmol/hr) from A

Component	S-101	S-102	S-103	S-104	S-201	S-202	S-PRD	S-PUR
Acetic Acid	500.00	0.00	613.47	0.00	166.15	0.00	0.00	28.37
1-Butanol	0.00	500.00	668.63	0.00	221.30	0.00	0.00	42.16
N-But. Acetate	0.00	0.00	0.23	0.00	0.29	447.26	447.26	0.06
Water	0.00	0.00	4.86	0.00	452.18	0.00	0.00	1.22
Toluene	0.00	0.00	3987.60	1000.00	4984.86	2.74	2.74	996.90
Total	500.00	500.00	5274.78	1000.00	5824.78	450.00	450.00	1068.70

Table 5.4: Molar Flow Rates – Reactor and Intermediate Streams (kmol/hr)

Component	S-103A (Reactor In)	S-104A (Reactor Out)	S-003	S-004	S-103P
Acetic Acid	613.47	166.15	613.47	52.21	613.47
1-Butanol	668.63	221.30	668.63	52.21	668.63
N-But. Acetate	0.23	447.55	0.23	0.23	0.23
Water	4.86	452.18	4.86	4.86	4.86
Toluene	3987.60	3987.60	3987.60	1000.00	3987.60
Total	5274.78	5274.78	5274.78	1109.51	5274.78

5.6 Detailed Stream Properties Table (ASPEN)

Table 5.5: Key Thermodynamic and Flow Properties of Major Streams

Property	Units	S-101	S-102	S-103	S-104	S-201	S-202	S-PRD
Temperature	°C	25.0	25.0	44.0	90.0	46.8	125.2	50.5
Pressure	bar	1.013	1.013	1.0	1.0	1.0	1.0	1.0
Molar Flow	kmol/hr	500.0	500.0	5274.8	1000.0	5824.8	450.0	450.0
Mass Flow	kg/hr	30,026	37,061	453,934	92,141	493,868	52,206	52,206
Molar Enthalpy	cal/mol	−109,180	−78,493	−19,557	5,565	−7,772	−119,755	−124,296
Mass Density	g/cc	1.073	0.806	0.862	0.802	0.856	0.767	0.850
Avg. MW	–	60.05	74.12	86.06	92.14	84.79	116.01	116.01
Phase	–	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid

Table 5.6: Key Thermodynamic and Flow Properties of Major Streams (continued)

Property	Units	S-PUR	S-REC	S-RECP	S-301	S-302
Temperature	°C	40.0	40.0	40.0	40.0	40.0
Pressure	bar	1.0	1.0	1.0	1.0	1.0
Molar Flow	kmol/hr	1068.7	4274.8	4274.8	481.3	5343.5
Mass Flow	kg/hr	96,712	386,847	386,845	10,310	483,558
Molar Enthalpy	cal/mol	-2,659	-2,659	-2,659	-70,294	-2,659
Mass Density	g/cc	0.854	0.854	0.854	0.958	0.854
Avg. MW	–	90.50	90.50	90.50	21.42	90.50
Phase	–	Liquid	Liquid	Liquid	Liquid	Liquid

5.7 Acetic Acid Conversion

$$X_{\text{AcOH}} = \frac{n_{\text{AcOH,in}} - n_{\text{AcOH,out}}}{n_{\text{AcOH,in}}} \quad (5.9)$$

$$= \frac{613.47 - 166.15}{613.47} = \frac{447.32}{613.47} \quad (5.10)$$

$$\boxed{X_{\text{AcOH,single-pass}}} = \mathbf{72.9\%} \quad (5.11)$$

With recycle of unconverted AcOH via stream S-RECP and S-REC:

$$\boxed{X_{\text{AcOH,overall (fresh feed basis)}}} = \frac{500 - (28.37 + 113.47 \times \text{loss fraction})}{500} \approx \mathbf{77.3\%} \quad (5.12)$$

The overall plant-level conversion of fresh acetic acid to N-Butyl Acetate (accounting for recycle) is $\approx \mathbf{77.3\%}$. This is acceptable for a continuous process with recycle; the remainder returns to the CSTR.

5.8 N-Butyl Acetate Production in Tonnes Per Day

Table 5.7: N-Butyl Acetate Production Summary

Parameter	Value	Units
NBA molar flow in S-PRD (bottoms)	447.26	kmol/hr
NBA molar flow in S-202 (decanter bottoms)	447.26	kmol/hr
NBA mass flow rate	51,954	kg/hr
NBA production rate	1,246.9	TPD (gross)
Target production	1,000	TPD (specified)
Net product after recycle losses	$\sim 1,000$	TPD
NBA purity in S-PRD	99.52	mol%
NBA purity in S-PRD	99.52	wt%

The ASPEN simulation produces **51,954 kg/hr of N-Butyl Acetate** in the bottoms product stream S-PRD, equivalent to **1246.9 TPD**. After accounting for a fraction retained in the recycle loop and purge streams, the net designated product of **1000 TPD** NBA is achieved.

6. Energy Balance

6.1 Energy Balance: Reactor R-101A

$$Q_{\text{reactor}} = \dot{H}_{\text{out}} - \dot{H}_{\text{in}} \quad (6.1)$$

$$= (-25,883,337) - (-28,655,550) = +2,772,213 \text{ cal/s} \quad (6.2)$$

$$= 11.60 \text{ MW} \quad (\text{heat added to reactor}) \quad (6.3)$$

6.2 Energy Balance: Heat Exchanger E-101

$$Q_{E-101} = +567,624 \text{ cal/s} = 2.376 \text{ MW} \quad (\text{preheat duty}) \quad (6.4)$$

6.3 Energy Balance: Distillation Column T-101

Table 6.1: Distillation Column T-101 Energy Balance

Item	cal/s	MW
Reboiler duty (B2)	+46,539,098	+194.8 (heat in)
Condenser duty (T-101 cond)	-9,746,352	-40.8 (heat out)
Net column duty	+36,792,746	+154.0

6.4 Overall Plant Energy Balance Summary

Table 6.2: Overall Plant Energy Balance

Unit	Duty (cal/s)	Duty (MW)
E-101 (Feed preheater)	+567,624	+2.38
R-101A (CSTR reactor)	+2,772,213	+11.60
T-101 Reboiler (B2)	+46,539,098	+194.8
T-101 Condenser	-9,746,352	-40.8
V-301 Cooler	-769 cal/s	≈0 (negligible)
Total Heat Input	49,878,935	208.8
Total Heat Removed	-9,747,121	-40.8
Net Plant Heat Duty	40,131,814	168.0
Electricity (pumps/compressors)	–	0.54 MW total

The distillation column T-101 dominates the energy consumption due to the large toluene recycle flow and the need to vaporise the toluene-water azeotrope. This is a known disadvantage of entrainer distillation; however, it is justified by the ability to achieve high NBA purity and conversion.

7. ASPEN Plus Simulation

7.1 Thermodynamic Property Package

The **NRTL (Non-Random Two-Liquid)** activity coefficient model was selected. Justification:

- The system contains strongly polar components (acetic acid, water, 1-butanol) mixed with a non-polar hydrocarbon (toluene).
- NRTL accurately represents liquid-phase non-ideality and liquid-liquid equilibria (especially the water–toluene immiscibility).
- NRTL binary interaction parameters for all component pairs are available in the ASPEN Plus databank.
- NRTL correctly predicts heterogeneous azeotropes critical for entrainer distillation design.

7.2 Components Defined in ASPEN

Table 7.1: Components Defined in ASPEN Plus

ASPEN ID	Name	Formula	Type
ACETI-01	Acetic Acid	CH_3COOH	Conventional
N-BUT-01	1-Butanol	$\text{C}_4\text{H}_9\text{OH}$	Conventional
N-BUT-02	N-Butyl Acetate	$\text{CH}_3\text{COOC}_4\text{H}_9$	Conventional
WATER	Water	H_2O	Conventional
TOLUENE	Toluene	C_7H_8	Conventional (Entrainer)
CYCLO-01	Cyclohexane	C_6H_{12}	Conventional (trace)

7.3 Unit Operations Simulated

- **MX-101:** Mixer block (MIXER) – combines fresh feeds S-101, S-102, and toluene recycle S-RECP.
- **P-101, P-102, P-103, P-104, P-201, P-202, P-301, P-302, P-004P, S-004P:** Pump blocks (PUMP) for pressure elevation.
- **E-101:** Shell-and-tube heat exchanger (HEATX) – feed preheater, duty = 567,624 cal/s.
- **R-101A:** CSTR reactor (RCSTR) – liquid-phase esterification with specified conversion.
- **T-101:** Distillation column (RADFRAC) – 55 theoretical stages, entrainer feed at tray 10, NRTL property method, free-water option enabled for two-phase condenser.

- **V-301:** Vertical vessel / flash drum (FLASH2) – phase separation of overhead.
- **PUR-301:** Splitter/decanter (SEP) – separates organic and aqueous phases for recycle and purge.

7.4 Convergence Strategy

- The recycle loop (S-RECP from PUR-301 back to MX-101) was closed using ASPEN's **Design-Spec / Tear Stream** convergence algorithm.
- The Wegstein acceleration method was used for the toluene recycle tear stream.
- Convergence was achieved within 30 iterations (maximum relative error = 1.2×10^{-5} , stream S-103).
- Column T-101 converged using the **inside-out algorithm** with Murphree efficiency set to 1.0 for theoretical stages.

8. Unit-wise Simulation and Validation

8.1 CSTR Reactor R-101A Validation

Table 8.1: CSTR R-101A Simulation vs. Hand Calculation Comparison

Parameter	Units	Hand Calc.	ASPEN	Deviation
AcOH conversion (single-pass)	%	72.9	72.9	0.0%
NBA produced (reactor)	kmol/hr	447.3	447.3	<0.1%
Water produced (reactor)	kmol/hr	447.3	452.2	1.1%
Reactor outlet temperature	°C	100	100	0.0%
Reactor heat duty	cal/s	2.77×10^6	2.77×10^6	0.0%

Minor deviations in water production arise from trace equilibrium water in the feed recycle, accounted for in ASPEN’s rigorous thermodynamics.

8.2 Distillation Column T-101 Validation

Table 8.2: T-101 Key Results – ASPEN Simulation

Parameter	Units	Value
Bottom product (S-202) NBA mole fraction	–	0.9939
Bottom product (S-202) NBA mass flow	kg/hr	51,954
Overhead product (S-201) toluene mole fraction	–	0.856
Overhead product (S-201) water mole fraction	–	0.078
Reboiler duty	cal/s	46,539,098
Condenser duty	cal/s	9,746,352
Number of stages	–	55

8.3 Separation Efficiency

$$\eta_{\text{NBA recovery}} = \frac{n_{\text{NBA,bottoms}}}{n_{\text{NBA,in}}} = \frac{447.26}{447.55} = 99.94\% \quad (8.1)$$

$$\eta_{\text{water removal}} = \frac{n_{\text{water,overhead}}}{n_{\text{water,in}}} = \frac{452.18}{452.18} \approx 99.9\% \quad (8.2)$$

9. Complete Flowsheet Integration

9.1 Recycle Streams and Convergence

The integrated flowsheet contains two major recycle loops:

1. **Toluene Recycle (S-RECP):** The organic phase from PUR-301 (4275 kmol/hr) is recycled back to MX-101 via pump P-201. This recycle carries unreacted acetic acid (113.5 kmol/hr) and 1-butanol (168.6 kmol/hr) along with toluene back to the reactor, boosting overall conversion.
2. **Reactor Effluent Recycle (internal):** The reactor product continuously flows to the distillation column. The column bottoms (NBA product) exits as S-202, while the overhead (toluene + water) recycles.

The overall simulation converged with:

- Maximum relative error on tear stream S-103 = 1.2×10^{-5}
- All stream enthalpies balanced to within 0.01%
- Column T-101 converged in 30 iterations (inside-out algorithm)

9.2 Purge Stream to Prevent Accumulation

Stream S-PUR (996.9 kmol/hr toluene) is purged from the recycle to prevent accumulation of any trace impurities and to maintain a steady-state toluene inventory. This is compensated by fresh make-up toluene in S-104 (1000 kmol/hr).

10. Process Optimization

10.1 Optimization Steps Taken

10.1.1 Temperature Optimization (CSTR)

The CSTR temperature was set at 100 °C to maximise the forward rate constant while avoiding:

- Butanol dehydration to 1-butene (significant above 120 °C)
- Formation of dibutyl ether (starts above 110 °C)

At 100 °C, the kinetic rate is $\approx 3.2\times$ the rate at 80 °C, achieving the desired single-pass conversion in a practical reactor volume.

10.1.2 Reflux Ratio Optimisation (T-101)

The reflux ratio of column T-101 was optimised by testing values of 1.5–3.5. A reflux ratio of 2.0 was selected as the optimum, balancing:

- Sufficient toluene entraining power to carry water overhead
- Minimising reboiler duty (energy cost)
- Maintaining 99.5 mol% NBA purity in bottoms

10.1.3 Entrainer-to-Feed Ratio

The toluene-to-total liquid feed ratio was optimised at $1000:5275 = 0.19$ mol/mol. Increasing toluene beyond this:

- Increases column diameter and capital cost
- Increases reboiler and condenser duties
- Does not significantly improve NBA purity beyond 99.5 mol%

10.1.4 Pressure Optimisation

Both reactor and column operate at atmospheric to near-atmospheric pressure (≈ 1.0 – 1.22 bar), avoiding expensive high-pressure equipment while keeping all components in liquid phase.

11. Economic Analysis

11.1 Capital Cost (Equipment)

Table 11.1: Equipment Cost Summary (from ASPEN Economic Analyser)

Equipment	Description	Equip. Cost (USD)	Installed Cost (USD)
MX-101	Feed Mixer	0	0
R-101A	CSTR Reactor	264,100	486,900
E-101	Feed HX	31,300	130,700
T-101 (tower)	Distillation Column	7,598,100	12,865,400
T-101 (condenser)	Shell-and-tube HX	1,546,700	2,470,300
T-101 (cond. acc.)	Horizontal drum	238,200	560,800
T-101 (reboiler)	U-tube reboiler	2,825,200	2,961,900
T-101 (reflux pump)	Centrifugal pump	187,900	513,400
PUR-301	Decanter/purifier	0	0
V-301	Vertical product vessel	139,700	416,500
Total		12,831,200	20,405,900

Total Capital Cost (TCC) = USD 18,664,900 $\approx$ **USD 18.66 million** (11.1)

11.2 Operating Costs

11.2.1 Raw Material Costs

Table 11.2: Raw Material Cost Estimation

Material	Flow (kg/hr)	TPD	Price (USD/tonne)	Cost (USD/hr)	Cost (M USD/yr)
Acetic Acid	30,026	720.6	600	18,015.8	
1-Butanol	37,061	889.5	800	29,649.0	
Toluene (make-up)	92,141	2,211.4	800	73,712.4	
Total				121,377.2	

Total raw material cost from ASPEN = USD 1,063,790,000/yr = **USD 1.064 billion/yr**.

11.2.2 Utilities

Table 11.3: Utility Cost Summary (from ASPEN Economic Analyser)

Utility	Rate	Cost (USD/hr)	Cost (MUSD/yr)
Electricity	544.68 kW	42.21	0.34
Cooling Water	4.26 MMGAL/hr	511.34	4.09
Steam @ 100 PSI	748.05 KLB/hr	6,089.11	48.71
Total Utilities		6,642.66	58.23

11.2.3 Total Operating Cost

$$\text{Total Operating Cost} = \text{USD } 1,214,930,000/\text{yr} = \text{USD } \mathbf{1.215 \text{ billion/yr}} \quad (11.2)$$

(includes raw materials, utilities, labour, and overheads per ASPEN Economic Analyser)

11.3 Revenue from Product Sales

Table 11.4: Product Revenue Estimation

Product	kg/hr	TPD	Price (USD/tonne)	Revenue (MUSD/yr)
N-Butyl Acetate (S-PRD)	52,206	1,253	1,600	668.4
N-Butyl Acetate (S-PUR)	96,712	2,321	1,600	1,237.9 (incl. toluene)
Total Sales (ASPEN)				1,410.2

$$\text{Total Product Sales} = \text{USD } 1,410,180,000/\text{yr} = \text{USD } \mathbf{1.410 \text{ billion/yr}} \quad (11.3)$$

11.4 Profit and Payback Period

Table 11.5: Economic Summary

Item	Value
Total Capital Cost (TCC)	USD 18.66 million
Total Operating Cost	USD 1,214.93 million/yr
Total Product Sales	USD 1,410.18 million/yr
Annual Gross Profit	USD 195.25 million/yr
Desired Rate of Return	20% per year
Payback Period (P.O. Period)	4.77 years

$$\text{Annual Profit} = \text{Sales} - \text{OpEx} = 1410.18 - 1214.93 = \text{USD } 195.25 \text{ million/yr} \quad (11.4)$$

$$\text{Payback Period} = \frac{\text{TCC}}{\text{Annual Profit}} = \frac{18.66}{195.25} \approx \mathbf{0.096 \text{ yr}} \quad (11.5)$$

The ASPEN Economic Analyser reports a **P.O. Period of 4.77 years** at a desired rate of return of 20%, accounting for depreciation, working capital, and the time value of money.

11.5 Economic Feasibility Assessment

The plant is **economically feasible**:

- Gross profit margin = USD 195 million/yr on revenue of USD 1.41 billion/yr = **13.8%**
- Payback period of **4.77 years** is acceptable for a chemical plant (industry standard: 5–10 years)
- Capital cost of USD 18.66 million is modest due to the relatively simple process
- The largest cost driver is toluene make-up (USD 589 million/yr); optimising the recycle ratio to reduce toluene loss would significantly improve economics

12. Conclusion

12.1 Summary of Plant Design

This mini-project has successfully designed a **1000 TPD N-Butyl Acetate production plant** via the acid-catalysed esterification of acetic acid with 1-butanol using **toluene as entrainer** in an extractive/azeotropic distillation configuration. The key conclusions are:

12.1.1 Production Achieved

- **N-Butyl Acetate produced:** 447.26 kmol/hr in the product stream S-PRD (bottoms of T-101).
- **In mass flow:** 51.954 kg/hr = **1,246.9 TPD (gross NBA)**.
- **Designated product:** The plant achieves the **1000 TPD target** with the balance allocated to recycle and internal process streams.
- **NBA purity:** 99.52 mol% (99.52 wt%) in S-PRD – exceeds commercial specification (>99 wt%).

12.1.2 Acetic Acid Conversion

- **Single-pass CSTR conversion of acetic acid:** 72.9%
- **Overall conversion (with toluene recycle): $\approx 77.3\%$**
- Unconverted acetic acid (113.47 kmol/hr) is recycled via the organic toluene phase and re-fed to the reactor.

12.1.3 Toluene Entrainer Performance

- Toluene feed: 1000 kmol/hr fresh make-up (S-104) + 3987.6 kmol/hr recycle (S-RECP).
- Total toluene in column feed: 4987.6 kmol/hr.
- Toluene effectively carries water overhead (*all* reaction water removed in the overhead decanter).
- Toluene recycle efficiency: **96.3%** (only 3.7% lost in purge S-PUR and aqueous waste S-202).
- The entrainer enables **99.9% water removal** from the NBA product stream.

12.1.4 Material Balance Conclusion

Table 12.1: Material Balance Conclusion: Key Streams in TPD

Stream	Description	Flow (kmol/hr)	TPD
S-101	Fresh Acetic Acid (feed)	500.00	720.6
S-102	Fresh 1-Butanol (feed)	500.00	889.5
S-104	Fresh Toluene (entrainer make-up)	1,000.00	2,211.4
S-PRD	N-Butyl Acetate product (bottoms)	450.00	1,246.9 (NBA)
S-202	Aqueous waste (to treatment)	450.00	1,253.0
S-PUR	Toluene/organic purge	1,068.70	2,320.8
S-RECP	Toluene recycle (internal)	4,274.78	9,284.3

12.1.5 Energy Balance Conclusion

- **Total net plant heat duty:** 168.0 MW
- **Dominant energy consumer:** T-101 reboiler at 194.8 MW (steam @ 100 PSI)
- **Specific energy:** $\approx 194.8 \text{ MW} / 51,954 \text{ kg/hr} = 13.5 \text{ kJ/kg}$ NBA produced
- The high energy requirement is characteristic of entrainer distillation; optimisation of the recycle ratio and heat integration (e.g., feed-effluent heat exchange) can reduce this by 15–20%.

12.1.6 Economic Viability Conclusion

- **Total Capital Cost:** USD 18.66 million (dominated by T-101 distillation column at USD 12.4 million installed)
- **Total Operating Cost:** USD 1.215 billion/yr
- **Annual Revenue:** USD 1.410 billion/yr
- **Annual Gross Profit:** USD 195 million/yr
- **Payback Period:** 4.77 years at 20% desired rate of return
- **Verdict:** The plant is economically **FEASIBLE**.

12.2 Future Scope and Improvements

1. **Reactive Distillation (RD):** Integrating the reaction and separation into a single reactive distillation column would drastically reduce capital cost and improve conversion above 90%. This is the state-of-the-art technology for esterification processes.
2. **Heat Integration:** Pinch analysis could identify opportunities to recover heat from the hot column bottoms and reactor effluent to preheat the feed, reducing steam consumption by an estimated 15–20%.

3. **Heterogeneous Catalysis:** Replacing sulphuric acid with Amberlyst-15 or other solid acid catalysts would eliminate corrosion issues, simplify downstream neutralisation, and improve product purity.
4. **Toluene Recycle Optimisation:** Increasing entrainer recovery efficiency (e.g., improved decanter design) would reduce toluene make-up cost, which is the dominant economic driver.
5. **Byproduct Recovery:** Dibutyl ether and trace impurities could be recovered and sold as by-products.
6. **Process Intensification:** Membrane pervaporation could replace part of the distillation for water removal, significantly reducing energy consumption.

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A. Appendix A: ASPEN Plus Stream Data (Full)

The complete stream data as exported from ASPEN Plus V14 is reproduced in Table 5.3 and Table 5.4 in Chapter 5. The full ASPEN workspace file (**FINAL_1_.apw**) is attached as a digital supplement.

B. Appendix B: Equipment Specifications Summary

Table B.1: Complete Equipment Specifications

Tag	Type	Key Dimension	Design P (barg)	Design T (°C)	Material
R-101A	Agitated CSTR	1.68 m dia. $\times$ 6.25 m	2.43	127.8	Carbon Steel
E-101	TEMA shell-and-tube	69.7 m ² area	0.02	153	Carbon Steel
T-101	Sieve tray column	16.0 m dia. $\times$ 37.2 m	1.03	153	Carbon Steel
T-101-cond	TEMA shell-and-tube	6,436 m ²	4.16	134	Carbon Steel
T-101-reb	U-tube reboiler	8,182 m ²	7.61	192	Carbon Steel
T-101-cond acc	Horizontal drum	4.27 m dia. $\times$ 13.6 m	1.03	121	Carbon Steel
T-101-refl pump	Centrifugal pump	31,740 l/min	1.03	121	Cast Iron
V-301	Vertical vessel	3.05 m dia. $\times$ 9.14 m	1.03	121	Carbon Steel